

STAT 20000 Class Meetings 2 & 3
Chapter 1-2 Experiments & Observational Studies

Explanatory Variable & Response

For each of the following claims, please identify the explanatory variable and the response

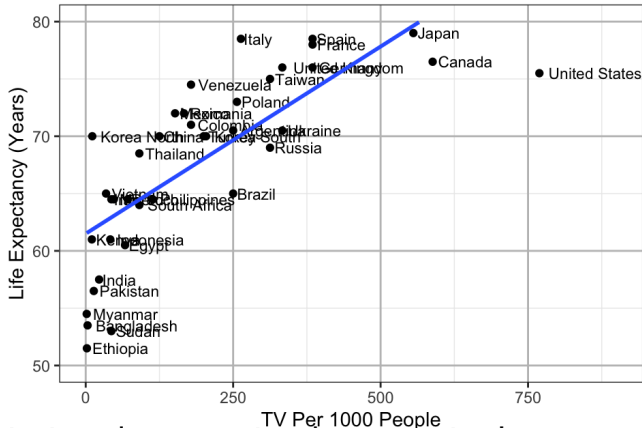
1. *"Daily Exercise Improves Mental Performance"*
 - ▶ Explanatory: Daily Exercise
 - ▶ Response: Mental Health
2. *"Want to lose weight? Watch more TV shows!"*
 - ▶ Explanatory: Watching More TV
 - ▶ Response: Weight Loss
3. *"More ice creams sales have been linked to more deaths by drowning"*
 - ▶ Ice Cream Sales
 - ▶ Count of Death by Drowning

Association and Causation

Two variables are **associated** if values of one variable tend to be related to values of the other variable.

Two variables have a **causal (or “cause-and-effect”) relation**, if changing the value of the explanatory variable influences the value of the response variable

Should We Buy More TVs to Live Longer?



Association does not imply causation!

Source: <http://jse.amstat.org/v2n2/datasets.rossman.html>

"Televisions, Physicians, and Life Expectancy" by Allan J. Rossman

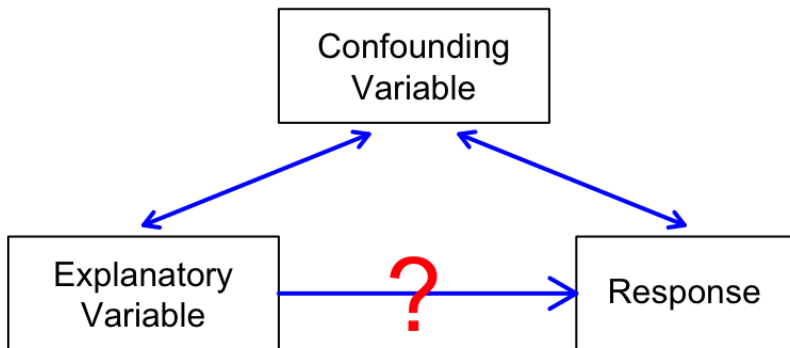
Journal of Statistics Education v.2, n.2 (1994)

Confounding Variable

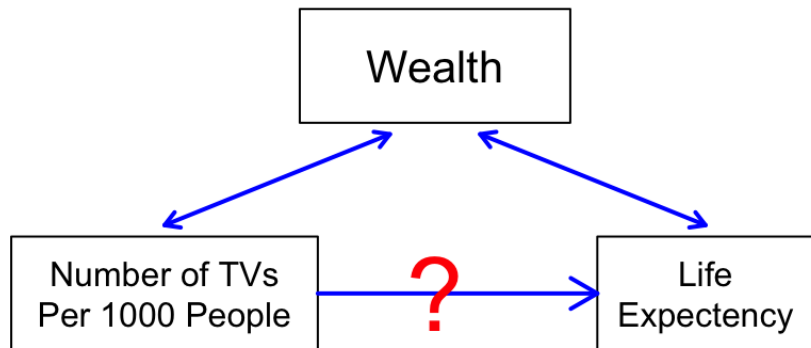
Definition: A third variable associated with both the explanatory variable and the response is called a *confounding variable*

- ▶ A confounding variable can offer a plausible explanation for an association between the explanatory and the response variables
- ▶ *Whenever confounding variables are (potentially) present, a causal relationship cannot be established*

Confounding Variable

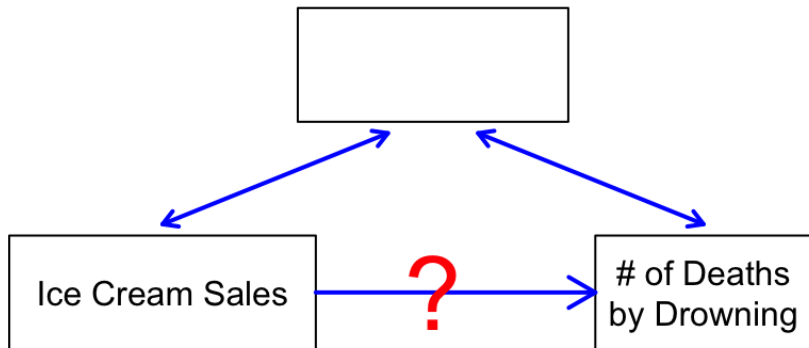


TVs and Life Expectancy



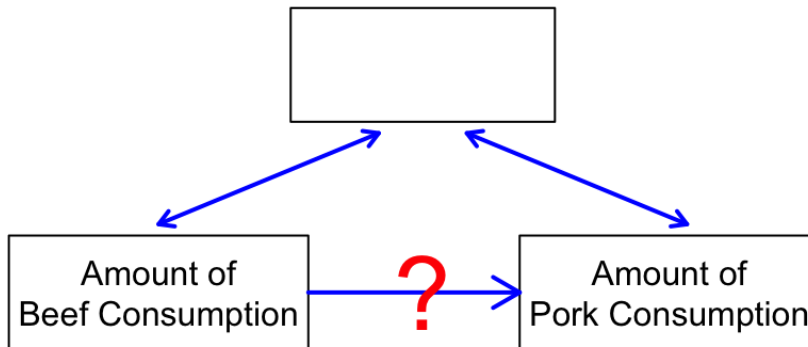
Identify a Confounding Variable (1)

Association: More ice creams sales have been linked to more deaths by drowning



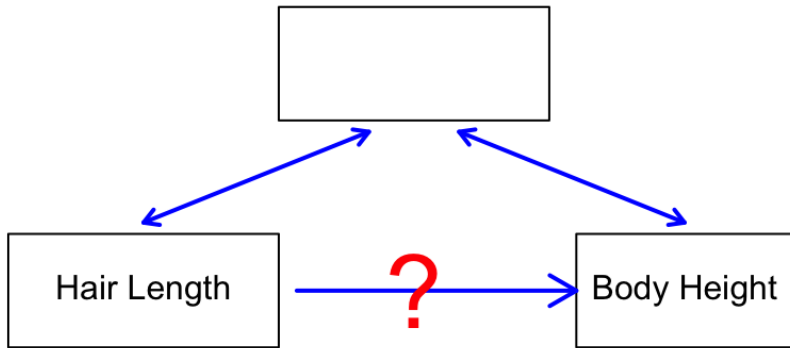
Identify a Confounding Variable (2)

Association: The total amount of beef consumed and the total amount of pork consumed worldwide are positively related over the past 100 years



Identify a Confounding Variable (3)

Association: People with shorter hair tend to be taller.



Experiment v.s. Observational Study

An **observational study** is a study in which the researcher does not actively control the value of any variable, but simply observes the value as they naturally exist.

An **experiment** is a study in which the researcher actively controls one or more of the explanatory variables

Observational Studies

- ▶ Many studies in social science and public health are observational studies.
- ▶ Example: to study the effect of smoking on health
 - ▶ Treatment group ("exposed group"): smokers;
 - ▶ Control group: no-smokers;
- ▶ Since the investigators have no control over allocation, the treatment group and control group might be different in important aspects other than the treatment, i.e.,
there are many confounding variables.

Observational studies only established **association**, NOT **causation**!

Randomization

How can we make sure to avoid confounding variables?

RANDOMLY assign
values of the
explanatory variable

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- ▶ For experiments, the explanatory variable is “treated or not”
- ▶ Randomly $\neq$ Haphazardly
- ▶ Avoid the human factor: use coin tossing/random number generation.

Controlled Experiment on Clofibrate (FPP, Ch 2, Sec 2)

	Clofibrate		Placebo	
	Number	Deaths	Number	Deaths
Total group	1103	20%	2798	21%

► Is Clofibrate effective?

Controlled Experiment on Clofibrate (FPP, Ch 2, Sec 2)

	Clofibrate		Placebo	
	Number	Deaths	Number	Deaths
Adherers	708	15%		
Non-adherers	357	25%		
Total group	1103	20%	2798	21%

- ▶ Is Clofibrate effective?
- ▶ Adherers v.s. non-adherers in the treatment group
Does this mean that Clofibrate is actually effective?

Controlled Experiment on Clofibrate (FPP, Ch 2, Sec 2)

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- ▶ Adherers v.s. non-adherers in the control group

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Does this mean that Clofibrate is actually effective?
- ▶ Adherers v.s. non-adherers in the control group
- ▶ Conclusion
 - ▶ Clofibrate has no effect;
 - ▶ Adherers are different from non-adherers;

Incentive for Smoke Quitting

Smokers at a big company were invited to participate in a smoking cessation program and randomly assigned to one of two groups.

- ▶ **Reward Only:** subjects got a cash award if stopped smoking for 6 months
- ▶ **Deposit:** subjects had to deposit some money first, could get deposit back + a bonus if stopped smoking.

Reward Only Group

	Quit Smoking	Not Quit	Total
Agreed to Participate	156	758	914
Declined to Participate	3	100	103
Total	159	858	1017

Deposit Group

	Quit Smoking	Not Quit	Total
Agreed to Participate	78	68	146
Declined to Participate	30	877	907
Total	108	945	1053

Which of the following is the correct and best conclusion of this experiment? Explain your choice.

- (a) “Reward” is more effective than “Deposit” since the success rate of the entire Reward-Only Group $159/1017 \approx 15.6\%$ is higher than that of the entire Deposit Group $108/1053 \approx 10.3\%$ and the two groups are divided by randomization.
- (b) “Deposit” seems more effective than “Deposit” since the success rates among those to agree to participate in the Deposit Group $78/146 \approx 53.4\%$ is higher than that for the Reward Group, $156/914 \approx 17.1\%$. However, this comparison cannot establish causality since people self-selected whether to participate in the assigned program.
- (c) Same comparison as (b) but we CAN make a causal conclusion that “Deposit” is better since this is a randomized experiment
- (d) Same comparison as (a) but we CANNOT make a causal conclusion that “Reward” is better due to the huge difference between the participation rates of the two groups.